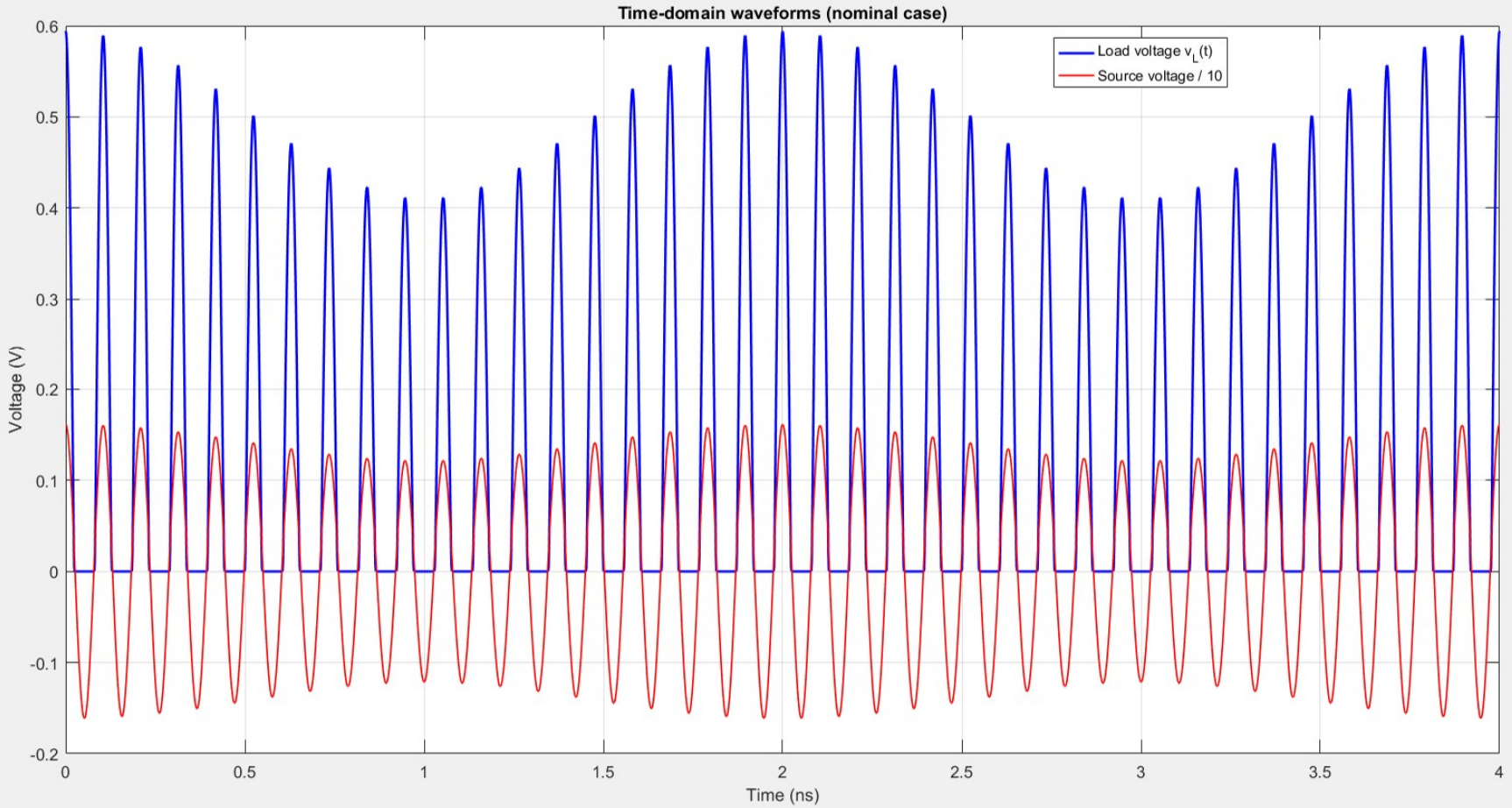
Figure 4

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exp06.m

```
179 –    xlabel('Frequency (GHz)'); ylabel(
180 –    title('Mixer output spectrum (RF =
181 –    legend('Location','northeast');
182
183 –    figure(2);
184 –    plot(PLO_sweep, CL_LO, 'bo-','Line
185 –        'DisplayName','Simulated CL');
186 –    plot([PLO_sweep(1) PLO_sweep(end)]
187 –    plot([PLO_sweep(1) PLO_sweep(end)]
188 –    grid on; xlabel('LO power (dBm)');
189 –    title(sprintf('Conversion loss ver
190 –    legend('Location','northeast');
191
192 –    figure(3);
193 –    plot(PRF_sweep, CL_RF, 'ro-','Line
194 –        'DisplayName','Simulated CL');
195 –    plot([PRF_sweep(1) PRF_sweep(end)]
196 –    plot([PRF_sweep(1) PRF_sweep(end)]
197 –    grid on; xlabel('RF input power (d
198 –    title(sprintf('Conversion loss ver
199 –    legend('Location','northwest');
200
201 –    figure(4);
202 –    tn = t*1e9; sel = tn <= 4;     % f
203 –    plot(tn(sel), vL_nom(sel), 'b-','L
204 –    plot(tn(sel), vs_nom(sel)/10, 'r-'
205 –    grid on; xlabel('Time (ns)'); ylab
206 –    title('Time-domain waveforms (nomi
207
208 –    fprintf('\nDone. Four figures gene
```

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```
================ PART A: FREQUENCY CONVERSION ================
Theoretical IF  (|fRF - fLO|) = 0.500 GHz
Theoretical sum (fRF + fLO)   = 19.500 GHz

--- Simulated spectrum lines (power into 50 ohm load) ---
Component          Theory (GHz)   Simulated (GHz) Deviation (%)  Power (dBm)
IF (difference)    0.5000         0.5000         4.768e-14      -21.015
LO                 9.5000         9.5000         0.000e+00      -2.648
RF                 10.0000        10.0000        0.000e+00      -18.158
Sum                19.5000        19.5000        0.000e+00      -21.015
FFT resolution = 50.0 MHz (coherent sampling, tones lie on bins)
Other products: 2*fLO (19.0 GHz) = -8.18 dBm, 2*fRF (20.0 GHz) = -49.25 dBm

================ PART B: CONVERSION LOSS ================
RF input power (available)         = -10.00 dBm
LO power (available)               = 7.00 dBm
IF output power at 0.50 GHz (sim)   = -21.015 dBm
Conversion loss CL = PRF - PIF     = 11.015 dB
Small-signal estimate: PIF = -20.990 dBm, CL = 10.990 dB
Difference (full sim - estimate)   = 0.025 dB
Sum-frequency power (sim)          = -21.015 dBm (sum-product conversion loss 11.015 dB)
Comparison with 6-10 dB range      : CL is above the typical 6-10 dB passive-mixer range (1.01 dB from upper limit)

--- CL versus LO power (PRF = -10 dBm) ---
PLO =   -5.0 dBm  ->  CL = 17.334 dB
PLO =   -3.0 dBm  ->  CL = 14.767 dB
PLO =   -1.0 dBm  ->  CL = 13.135 dB
PLO =    1.0 dBm  ->  CL = 12.193 dB
PLO =    3.0 dBm  ->  CL = 11.621 dB
PLO =    5.0 dBm  ->  CL = 11.256 dB
```

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PLO =    5.0 dBm  ->  CL = 11.256 dB
PLO =    7.0 dBm  ->  CL = 11.015 dB
PLO =    9.0 dBm  ->  CL = 10.852 dB
PLO =   11.0 dBm  ->  CL = 10.739 dB
PLO =   13.0 dBm  ->  CL = 10.664 dB

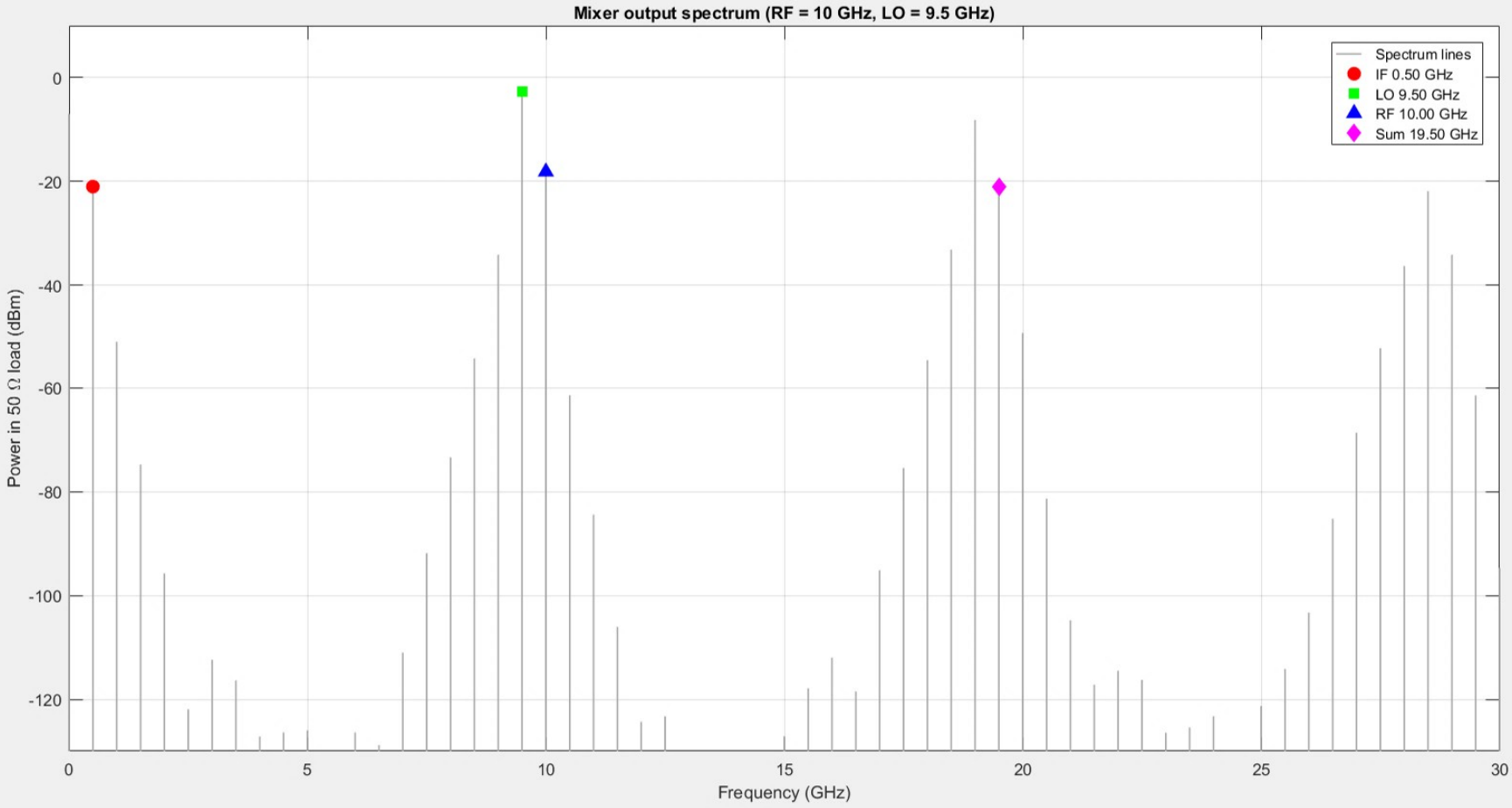
--- CL versus RF power (PLO = 7 dBm) ---
PRF =  -40.0 dBm  ->  CL = 10.990 dB
PRF =  -35.0 dBm  ->  CL = 10.990 dB
PRF =  -30.0 dBm  ->  CL = 10.990 dB
PRF =  -25.0 dBm  ->  CL = 10.991 dB
PRF =  -20.0 dBm  ->  CL = 10.993 dB
PRF =  -15.0 dBm  ->  CL = 10.998 dB
PRF =  -10.0 dBm  ->  CL = 11.015 dB
PRF =   -5.0 dBm  ->  CL = 11.069 dB
PRF =    0.0 dBm  ->  CL = 11.250 dB
PRF =    5.0 dBm  ->  CL = 11.932 dB

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Figure 1

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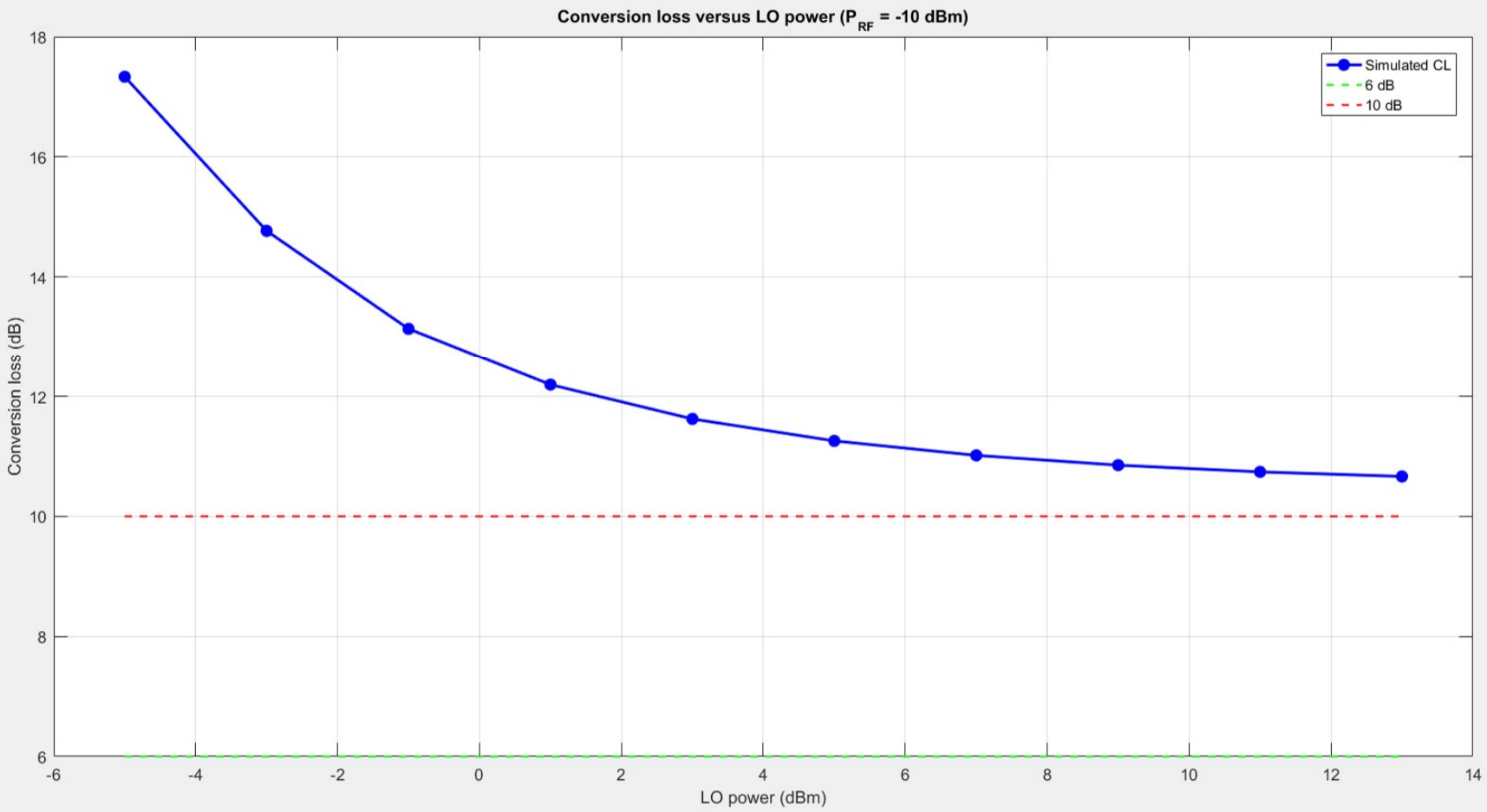
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Figure 2

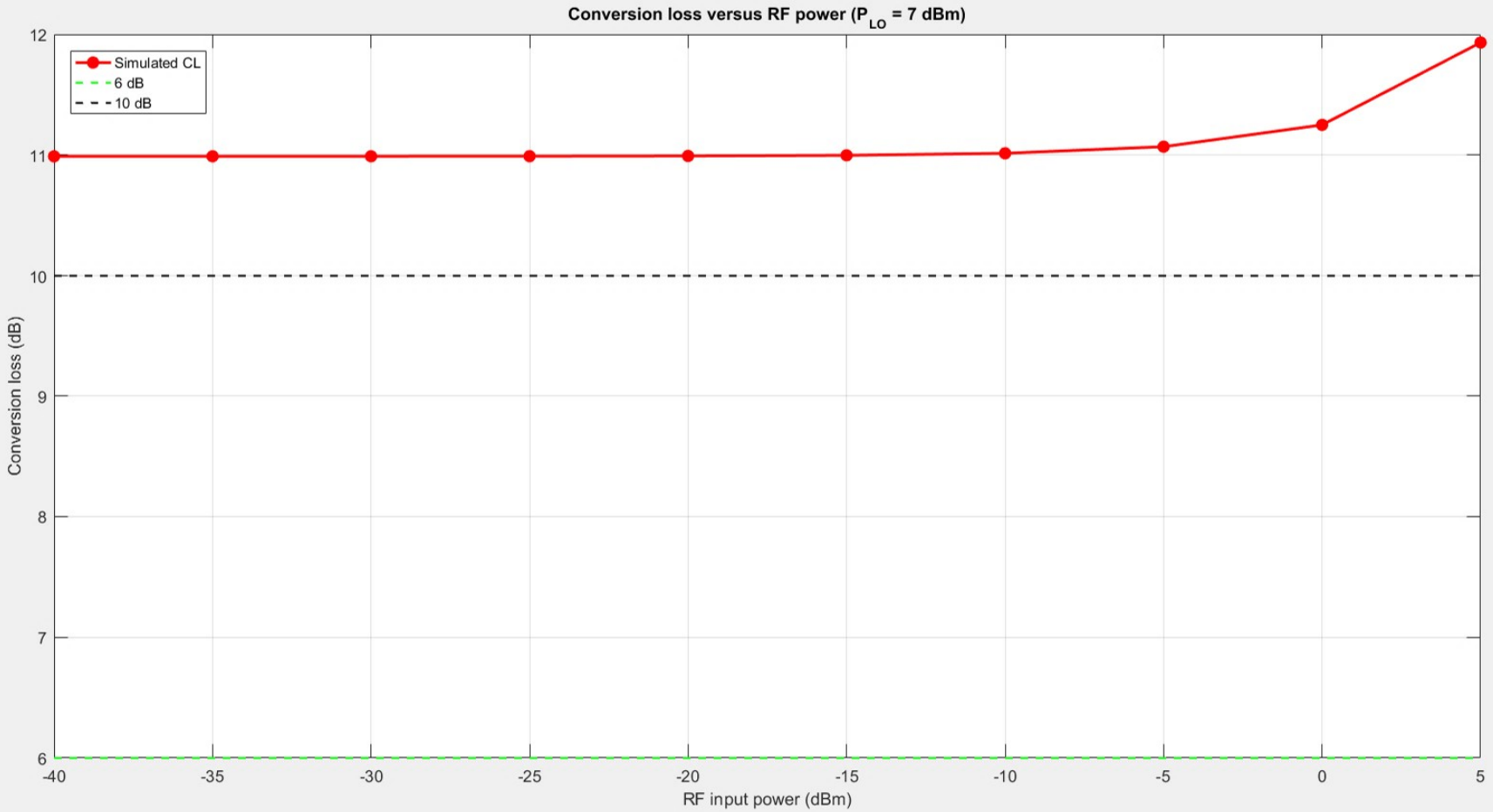
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